

Subject:

Request for Technical Critique / arXiv Sponsorship — “Geometric Foundations for Unified Physics”

Dear Professor Carlip,

My name is Jed Kircher. It's been more than 15 years since I attended UC Davis—though in labor relations, not physics. I'm a retired firefighter from Sacramento City Fire, and served as Director and Municipal VP for IAFF Local 522 (circa 2004–2010). I hope this helps explain my unconventional approach.

I'm reaching out with a framework that I believe may offer new insight into unifying general relativity and quantum mechanics. The enclosed executive summary, “Geometric Foundations for Unified Physics,” presents a geometric model where three foundational constraints—Zero (collapse), Infinity (expansion), and Chance (probability)—emerge as structural dimensions alongside space and time. This emergent order is not simply an extension, but a minimal completion of the metric, providing a bridge between relativity and quantum theory.

By encoding these constraints directly into the metric, the model appears to resolve singularities, geometrize quantum uncertainty, and predict measurable, falsifiable deviations in gravitational wave polarization and quantum interference.

The framework reduces cleanly to standard GR in the appropriate limits, and proposes several falsifiable consequences relevant to current experiments and observations.

Would you be willing to provide a technical critique, or (if merited) consider sponsoring the paper for arXiv submission? I realize my background is unconventional, but I am eager for rigorous criticism from an expert in the field. I am also reaching out to your colleague, Prof. Nancy Aggarwal, with a tailored note regarding experimental relativity and black holes, in hopes it may be of interest to her research.

I've attached a 2-page executive summary, and can provide the full paper or further details on request. Thank you very much for your consideration.

With respect,
Jed Kircher

Contact:

916-396-8071 (U.S.—this always gets through)

+420 728 77 44 94 (Czech Republic—current location)

1570 12th Ave
Sacramento, CA 95818

Abstract:

This paper presents a geometric framework for unified physics in which three foundational constraints—Zero (collapse), Infinity (expansion), and Chance (probability)—emerge as structural dimensions alongside the familiar axes of space and time. These emergent dimensions encode the boundaries of collapse, expansion, and probabilistic behavior directly into the metric, providing a natural completion to general relativity and a bridge to quantum mechanics. The resulting equations resolve singularities, geometrize quantum uncertainty, and explain cosmic acceleration without recourse to ad hoc parameters. The model predicts testable deviations in gravitational wave polarization, quantum interference, and vacuum fluctuation profiles, offering a falsifiable path to quantum gravity grounded in geometric necessity.

Draft: Cover Email to Prof. Nancy Aggarwal (UC Davis)

Subject:

Experimental Implications & arXiv Support – “Geometric Foundations for Unified Physics”

Dear Professor Aggarwal,

My name is Jed Kircher. It's been over 15 years since I attended UC Davis—though in labor relations, not physics. I'm a retired firefighter from Sacramento City Fire and previously served as Director and Municipal VP for IAFF Local 522. I hope you'll forgive the unconventional outreach.

I am writing to share a geometric framework that may offer new, testable predictions in experimental relativity and black hole physics. The enclosed executive summary, “Geometric Foundations for Unified Physics,” presents a model where three emergent constraints—Zero (collapse), Infinity (expansion), and Chance (probability)—act as structural dimensions, alongside space and time.

This approach encodes boundaries on collapse, expansion, and quantum uncertainty directly into the metric, predicting measurable, falsifiable deviations in gravitational wave signatures, black hole structure, and cosmic microwave background anisotropies. Several of these predictions could be relevant to current and upcoming gravitational wave and precision measurement experiments.

I realize my background is non-traditional, but my intent is to invite rigorous criticism and dialogue. If you find any of these experimental consequences potentially relevant to

your work, I would greatly value your feedback—or, if appropriate, your support as an arXiv sponsor.

I've attached a 2-page executive summary, and can provide the full paper or additional details on request. Thank you very much for your time and consideration.

I should mention I am also reaching out to your colleague, Prof. Stevin Carlip, with a technical summary focused on theoretical aspects of the framework. I hope you don't mind the parallel outreach—I believe both the experimental and theoretical perspectives are important for evaluating new ideas like this.

With respect,

Jed Kircher

Contact:

916-396-8071 (U.S.—this always gets through)

+420 728 77 44 94 (Czech Republic—current location)

1570 12th Ave

Sacramento, CA 95818